

# AUTONOMOUS SPIDER ROBOT

*Hexapod Locomotion with Servo Control & Obstacle Avoidance*

## Team Electronauts

Dept. of Electrical & Electronics Engineering

SSM Institute of Engineering and Technology

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# PROJECT OVERVIEW

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## What is it?

A quadruped walking robot inspired by spider locomotion, built with 4 legs (2 servos each = 8 total), capable of multi-directional movement on varied terrain.



## Current Phase

Functional prototype using Arduino Nano V3, 8 SG90 servo motors, PCA9685 servo driver, and HC-SR04 ultrasonic sensor for basic obstacle detection.



## Goal

Scale up to Raspberry Pi or NVIDIA Jetson Nano for full autonomous navigation, computer vision-based pathfinding, and real-time terrain adaptation.

# HARDWARE COMPONENTS

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01

## Arduino Nano V3

*Main Microcontroller*

ATmega328P, 16MHz, 32KB Flash — Brain of the robot

02

## PCA9685 Servo Driver

*16-Channel PWM Controller*

I2C interface, controls all 8 servos with precise PWM

03

## 8× SG90 Servo Motors

*Leg Actuators*

2 servos per leg × 4 legs — enables coxa & femur motion

04

## HC-SR04 Ultrasonic

*Obstacle Detection Sensor*

Range: 2cm–400cm, triggers avoidance maneuver

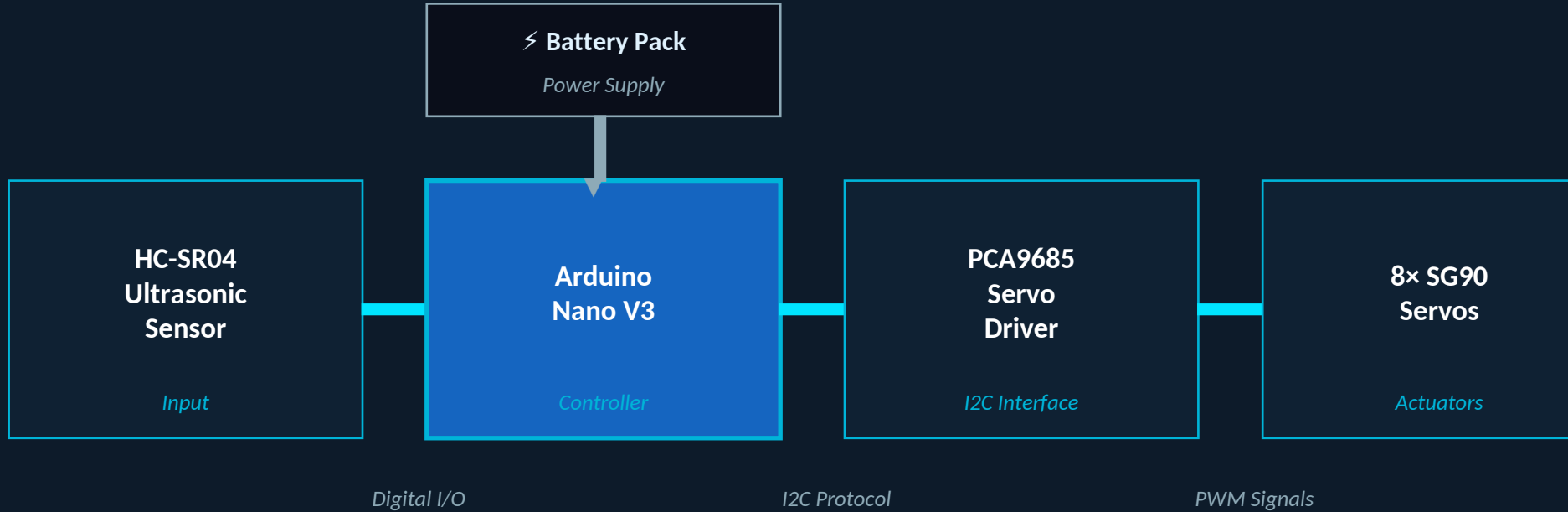
05

## Li-Po Battery Pack

*Power Supply*

Portable power for motors and logic circuits

# SYSTEM ARCHITECTURE



# WORKING PRINCIPLE

## Leg Mechanism

- 4 legs, each with 2 DOF (Degrees of Freedom)
- Coxa servo: horizontal swing (forward/back)
- Femur servo: vertical lift (up/down)
- Alternating tripod or wave gait pattern
- Coordinated sequence ensures stability
- Capable of turning, forward & reverse motion



## Obstacle Avoidance Logic

- 1 HC-SR04 emits ultrasonic pulse
- 2 Echo return time → distance calc
- 3 If distance < threshold → STOP
- 4 Turn left or right to avoid
- 5 Resume forward movement

# CURRENT PROTOTYPE — SPECS AT A GLANCE

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**8**

**Servo Motors**

*SG90, 2 per leg*

**4**

**Legs**

*Quadruped design*

**2 DOF**

**Per Leg**

*Coxa + Femur*

**I2C**

**Communication**

*Arduino ↔ PCA9685*

**400cm**

**Sensor Range**

*HC-SR04 max range*

**Nano V3**

**Microcontroller**

*ATmega328P core*

# FUTURE SCOPE & UPGRADES

## Phase 1 (Current)

- Arduino Nano V3 control
- 8 servo motor prototype
- Basic obstacle avoidance
- Manual / pre-programmed gait



## Phase 2 (Near Term)

- Upgrade to Raspberry Pi 4
- Add camera module (Pi Cam)
- Python-based autonomous nav
- Wi-Fi remote monitoring



## Phase 3 (Advanced)

- NVIDIA Jetson Nano / Orin
- Computer Vision + SLAM
- Real-time terrain mapping
- AI-powered path planning

# MEET TEAM ELECTRONAUTS

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